Doctoral Student Experimenting Lab:
Explore the Responsible Use of AI in Generating Scientific Texts, Images, Audio and Code

University of Zurich | ETH zürich | University of Basel

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Course Syllabus

Doctoral Student Experimenting Lab (Block Course):
Explore the Responsible Use of AI in Generating Scientific Texts, Images, Audio and Code (4 days, 1 ECTS)

Lecturers: Melanie Paschke (Zurich-Basel Plant Science Center), Jeanine Reutemann (ETH), Thomas Frei (Student Project House), Réka Mihálka (ETH Library), Miklos Frank (ETH Library), Paulina Zybinska (ZHAW), Alexis Shakas (ETH), Franziska Oschmann (ETH Research IT Platform), Fabian Schmid (ETH Library)& Robin Staab (ETH)

Tutors: Borges Goncalves Daniel (ETH), Onishchuk Daria (ETH)

Coordinator: Bojan Gujas (ETH, PSC)

Location: ETH Zurich Centre, main building

Dates & Rooms: Monday, **May 5th** (ETH, HG F26.3),
Tuesday, **May 6th** (morning:Student Project House in Clausiusstrasse 16 /
Tuesday, **May 6th** (afternoon: ETH, HG F26.3),
Tuesday, **May 13th** (Media & Methods Lab ETH RZ D21 in Clausiusstrasse 59),
Wednesday, **May 14th** (ETH, HG F26.3).

Time: 9:00 – 18:00

Credit Points: 1 ECTS

Course Contents:
Interested in using Generative AI in your scientific work processes in a responsible and ethical manner? This block course for PhD students allows for experimenting with generative AI to generate texts, images, and audio that can be used in various scientific contexts, from presentations to publications. PhD students are invited to engage in a problem-based learning setting, tackling concerns and critical topics related to the use of AI-based tools with research data, private data, and output generation, including images, text, short videos, audio, and code. Invited experts will share their knowledge in several hands-on workshops, covering topics such as the generation of scientific illustrations and the customization of AI models in GPTs and bots.

Learning objectives of the labs, in general:
- Hands-on training on using generative AI in a responsible, efficient and professional way in daily scientific work
- Competencies to deal with limitations, potential pitfalls and benefits of AI-based tools in relation to research integrity and ethics.
- Experiences about generative AI through experimenting with AI.

Targeted group: PhD students of ETH Zurich, University of Zurich and University of Basel.

Individual Performance and Assessment: Ungraded semester performance. Students will actively participate during the course with group presentations at the end of each block and a documentation to be handed in.

Moodle: https://moodle-app2.let.ethz.ch/course/view.php?id=25422 (please login with your ETHZ credentials). If you never used Moodle, you will be prompted to register. Once registered, please send email to the coordinator @ bojan.gujas@usys.ethz.ch to be added to the course.

May 5, 2025
DAY 1: Large Language Models: Customization and Prompting.

ETH Zurich Centre, HG F26.3

Lecturers
Réka Mihalka, Melanie Paschke

Learning Objectives
After the completion of the course day, participants will be able

- Select suitable Large Language Models (LLMs) for their use cases.
- Understand the approaches that LLMs and reasoning LLMs use.
- Customize their ChatGPT or Gem environment.
- Set up a custom GPT or Gem for a wide range of use cases.
- Apply basic prompting techniques as chain-of-thought prompting or exemplars and understand how they are implemented in reasoning LLMs.
- Understand limitation in LLMs for bias, overconfidence or hallucinations in Large Language Models.
- Understand and apply mitigation strategies for scientific integrity.
- Can use gen AI and hold up the principles of scientific integrity.

Course Format
lecture, hands-on workshops, discussion groups, case studies

Lecture: Introduction to LLMs and prompting. This lecture provides an overview of currently available LLMs and the LLMs licenced by ETH Zurich. It also highlights key features of different platforms such as customization options and special functionalities (e.g., Canvas, deep research). The talk also covers basic prompting techniques (e.g., role, goal, context, examples, output requests) and more advanced approaches such as retrieval-augmented generation (RAG) and chain-of-thought prompting. Participants will be able to implement these concepts in the subsequent workshop.

Workshop – Develop a custom bot. As a practical continuation of the lecture, this workshop will help participants get started with the creation of their own custom tools. Participants will identify use cases, plan, prompt, and test their tools and gather feedback from other participants and the facilitator of the workshop. Requirements: Please bring your own laptop and set up a ChatGPT, Copilot, or Gemini account.

Workshops: Introduction to Limitations of LLMs and strategies for mitigation

LLMs have limitations: Bias happens if LLMs are amplifying certain stereotypes e.g. about a group, a person or is over-representing certain views, opinions or is excluding diversity, often inherited from the training data set. Explore the “Explorables Collection” of Google and see biases and stereotypes hidden in LLMs. Another limitation are hallucinations that arise through inference, sycophancy or “hay in the stack” behavior.

Strategies that you will try out to mitigate these limitations based on case studies are: e.g. Prefix prompting, Self-Correction, Explainable prompts, Conversational socratic dialogue, According to… prompting, Chain-of-Verification Prompting, Meta Prompting, Self-Consistency, Culturally sensitive prompts, Rich Semantic Details e.g. in image prompting. Create outputs that will represent diversity of cultures, societies, individuals and opinions and evidence.

Day Schedule

Time Slot	Lecture/Workshop Title	Key Topics Covered	Responsible
9:00 – 9:15	Welcome, introduction and syllabus of the course	Introduction of Participants	Bojan Gujas
9:15 – 10.30	**Lecture:** Introduction to LLMs, Customizing LLMs, Different LLMs for different purposes. Prompting strategies	Retrieval-augmented generation, chain-of-thought prompting	Réka Mihalka
10.30 – 10.45	**Break**		
10.45 – 12.30	**Workshop**: Develop your own custom tools		Réka Mihalka
12:30-13:30	Lunch Break		
13:30-14:45	**Workshop:** Bias and stereotypes in LLMs.	Inherent limitations of LLMs	Melanie Paschke
14:45-15:00	Break		
15:00-16:00	Hallucinations and other limitations of LLMs. Advanced prompting techniques to avoid limitations. Experimenting lab and case studies.	Inherent limitations of LLMs: Case study work and techniques to overcome group bias, overrepresentation of data, overconfidence, hallucinations, sycophancy	Melanie Paschke
16:00-17:00	**Homework:** Current challenges in research integrity through AI: Start Case Study Work	See Material and Resources	Melanie Paschke

Materials and Resources:

a. Reading Materials (Preparatory reading, Course Materials, Course Handouts):

Melanie Paschke, Réka Mihálka and Manuel Sudau (2024). Cases for Research Integrity: Generative AI. Zurich Basel Plant Science Center: ETH Zurich. DOI: 10.3929/ethz-b-000664648

Paschke, Melanie, with contributions by: Petterson, Alexander, Mihalka, Réka and Manuel Sudau (2024). Teaching Collection: Exercises and hands-on examples for ethical use of generative AI. Zurich Basel Plant Science Center: ETH Zurich. https://doi.org/10.3929/ethz-b-000672206

b. Tools/Software/Datasets Required/Accounts created:
Bring your own laptop with an account to an LLM.

- You can use the ETHZ Copilot access to GPT-4 and DALL-E 3. Use your ETHZ-credentials to access Copilot.microsoft.com.
- You can obtain an ETH license to Gemini from the IT Shop: the approval process only takes a few minutes. Go to “Service Catalog” on the left-hand menu ➔ “Cloud Subscription” (on the right) ➔ “Request Cloud Subscription (non-Microsoft)” ➔ click through the remaining pages
- Imagen 3 in Gemini is also available via ETHZ login.

HOMEWORK for DAY 3 and DAY 4!!!

Discuss the following cases and present the outcomes as a short guideline (3 slides on day 3 and 4, first hour, see program): What is the challenge, the problem? How to deal with the situation? ETH or international recommendations or regulations?)

All cases are available in the Moodle: https://moodle-app2.let.ethz.ch/course/section.php?id=217371

For each case we accept 4 participants:

- [Confidentiality, generative AI] Compromising confidentiality in the peer review process
- [Generative AI]: The environmental footprint of generative AI
- [Peer Review, generative AI] R.I.P. scientific publishing and peer-review system
- [Authorship, generative AI]: The notions of scientific authorship in the era of generative AI
- [Images, generative AI]: Image Editing and gen AI
- [Images, generative AI] The message of scientific content - Gen-AI images on posters and in graphical abstracts
- [Transparency, generative AI] Did an AI write this?
- [Fair Use, Copyright, generative AI] Data Feasts and Fair Use: AI's Legal Appetite Under Scrutiny
- [Human in the loop, generative AI] The Human Paradox: Objective Instruments, Subjective Minds - Revealing the Illusion of Precision: Between Machine Accuracy and Human Judgment
- [Automation in research, parroting, epistemic risks, generative AI] Automation of the research cycle: epidemic opportunity or risk?

Sign for the Group Work Here:
https://docs.google.com/spreadsheets/d/1r0jaMzSjr5QwQiSz2dko6poyrRjMMqdwbd5Ngc7sibY/edit?usp=sharing

May 6th, 2025

DAY 2: AI-supported scientific workflows

ETH Zurich Centre, Morning 09:00-12:00: Student Project House in Clausiusstrasse 16
Afternoon 13:00-18:00 ETH, HG F26.3

Lecturers & Tutors

Thomas Frei (ETH), Alexis Shakas (ETH), Alessandra Ruggia (ETH Library), Raymond Grenacher (ETH Library), Miklos Frank (ETH Library)

Day Abstract

This course day offers a hands-on exploration of generative AI tools and their practical applications in scientific research. We will begin with a **"Chat with PDF"** workshop, using it as a case study to demystify how Large Language Models (LLMs) operate from a machine learning perspective, and to demonstrate how such models can be loaded and used in a Python environment. Next, the **AI-assisted coding workshop** will focus on practical strategies for problem-solving when codding, showing participants how generative AI can support programming tasks.
Shifting focus to the essential soft skills of a modern researcher, the **ETH Library team** will present a range of AI-powered tools designed to streamline and improve **literature research**, helping participants stay up to date and work more efficiently. Last workshop will provide insights into various useful applications of AI in scientific work.

Course Format

lecture, workshops, hands-on applications of visual genAI

Day Schedule

Time Slot	Lecture/Workshop Title	Key Topics Covered	Responsible
09:00-12:00	**Workshop 1:** Chat with your PDF	Working with LLMs in Python	Thomas Frei
12:00-13:00	Lunch Break		
13:00-15:00	**Workshop 2**: AI Assisted Coding in Python	Problem Solving in Python Programming	Alexis Shakas
15:00-15:15	Break		
15:15-17:00	**Workshop 3:** AI Toolbox for Literature Research	Bibliography	Miklos Frank, Alessandra Ruggia & Raymond Grenacher
17:00-18:00	**Workshop 4:** Chat GPT Upload Option	Tables & Photo Recognition	Bojan Gujas

Learning Objectives

After the completion of the course day, participants will be able
- to understand how LLMs work from ML perspective
- to code with AI assistance
- to create a bibliography with AI

Workshop Details

Workshop 1 – Chat with your PDF. You will learn how Large Language Models (LLMs) work, how you can use them, and what their limitations are. Explore the fusion of Computer Vision techniques with LLMs to extract structured information from PDF files effectively. Through hands-on exercises, participants will learn to develop a comprehensive program capable of analyzing PDF documents, extracting relevant data, and providing insightful answers to specific inquiries. Our goal is to give an all-in-one program that analyze your PDFs and answer specific questions about them. Requirements: No coding expertise in a specific language is required but a basic understanding of programming will help. Bring your own laptop.

Workshop 2 - AI Assisted Coding in Python. This workshop is targeted to anyone who wants to, or needs to write code to analyze their data, visualize results, simulate processes, or simply for the fun of it. We will start by learning two techniques to address problem solving, that allow us to design programs at a high-level (as text, or prompts) and also develop methods to test our hypotheses. This will serve as a basis for using AI assistants to write python code that will help solve our problems. We will address a toy problem together and then split into groups, or as individuals, to tackle separate problems and create code for them. You are also encouraged to "bring in your own problems" and work with them, but please follow the questionnaire supplied if you do so. Requirements: Bring your own laptops with already installed Visual Studio Code (https://code.visualstudio.com/) that is a powerful editor for python. Additionally, please download and install Python (https://www.python.org/downloads/) with at least the following packages; numpy, matplotlib, and scipy. For AI assisted code creation on VS code you can use Github coPilot (request and create a free account through ETH) but we can also prompt chat GPT .

Workshop 3 – AI Toolbox for Literature Research. This 2-hour workshop equips the participants with the knowledge to leverage AI-assisted tools for more efficient and effective literature research. We will explore the capabilities of current AI technologies designed to tackle challenges like information overload and identifying research connections.
The session covers practical applications of assisted bibliographic database search and discovery using citation networks. Through demonstrations of tools licensed at ETH (Scopus AI, Scite, Connected Papers, Research Rabbit), we will look at these functionalities in action. The workshop emphasizes critical evaluation, addressing AI limitations, and potential biases. Learn strategies to responsibly integrate these assistants into your existing research workflow, enhancing productivity while upholding academic integrity.

Workshop 4 – During this workshop we will explore several workflows to leverage generative AI in everyday scientific work.

May 13th, 2025

DAY 3: Responsible Image Creation & Scientific Visualisations

ETH Zurich Centre, Media & Methods Lab ETH RZ D21 in Clausiusstrasse 59

Lecturers & Tutors

Dr. Jeanine Reutemann, Paulina Zybinska, Daniel Goncalves, Daria Onishchuk

Abstract

On this day, we will explore GenAI with a focus on visual/audiovisual tools and their applications in research and communication. The course combines theoretical and practical, hands-on workshops, enabling PhD students to critically engage with cutting-edge technologies such as text-to-image, text-to-video, and image-to-video. Participants will work with their own research topic to create visual outputs such as graphical abstracts, infographics, avatars or potentially deepfakes.

Course Format

lecture, workshops, experimental audiovisual laboratory, hands-on applications of visual genAI

Day Schedule

Time Slot	Lecture/Workshop Title	Key Topics Covered	Responsible
09:00 – 09:40	**Presentation of Student Case Studies on Research Integrity**	Image Editing and gen AI (10 min) The message of scientific content - Gen-AI images on posters and in graphical abstracts (10 min) The environmental footprint of generative AI (10 min) Did an AI write this? (10 min)	MP, BG
09:40-10:30	**Lecture 1:** Visualizations in Science- GenAI as a potential paradigm shift in scientific endeavors?	Scientific visualizations, visuals in science, historical perspective to GenAI, visual abstracts, infographics, video abstracts, social media, science communication	JR
10:30-10:50	**Workshop 1:** Exercises with visual GenAI	First fast experimental tests with tools and tasks	JR, PZ, DG, DO
10:50-11:00	Break		
11:00-12:30	**Workshop 2:** create visuals for your own PhD topics with visual GenAI tools	text-to-image, sketch-to-image, inpainting etc.	JR, PZ, DG, DO
12:30-13:30	Lunch Break		
13:30-14:15	**Lecture 2:** Embrace the Fake – Using Deepfakes to Streamline Research Communication	digital twins, voice cloning, summary, automated video creation	PZ
14:15-16:00	**Workshop 3:** audiovisual GenAI generations with their own PhD topics	Image-to-video, avatar, deepfakes, voice-cloning	PZ, DG, DO, JR
16:00-17:00	**Plenary:** short 1-min pitches of the students GenAI experimental tests	Pitches, overview of productions	JR, PZ

Learning Objectives

- get introduced to the broad spectrum of visual and audiovisual GenAI tools
- learn about the vast potentials of visuals for research communication.

- experiment with audiovisual AI tools to create e.g. text-to-image, sketch-to-image, image-to-video and text-to-video outputs
- apply visual genAI tools to generate their own visual content
- work collaboratively to explore different visual genAI tools.
- present their own visual designs.

Materials and Resources
a. Reading Materials (Preparatory reading, Course Materials, Course Handouts):

b. Tools/Software/Datasets Required/Accounts created:
Preliminary decision is that workshop parts will be on Leonardo.AI, Luma.AI and own built AI. Info will be provided on the course day.

c. Any Additional Resources Required/Provided from/to Students:
- Bring your own (if possible fast) laptops; phones, drawing pads, sketchbooks
- if your own research involves visual material: bring them into the session, e.g. digital pictures, drawings, video documentation, visual field data etc. – or even real 3D-objects

Ethical Considerations & Responsible Use
a. How will ethical considerations and responsible AI use be incorporated into your session?
- will be addressed with historical and contemporary case study examples
b. Scenario examples or case studies used (if applicable):
- political propaganda, social media, science-society interface (tbd.)

May 14th, 2025

DAY 4: Privacy in LLMs

ETH Zurich Centre, HG F26.3

Lecturer

Franziska Oschmann (ETH Research IT Platform), Fabian Schmid (ETH Library) & Robin Staab

Day Abstract

This day we will explore the intersection of privacy regulations and machine learning, with a specific focus on privacy in long language models. Beginning with an introduction to global privacy regulations, participants will gain insights into emerging legislative frameworks and their implications for AI development. In workshops, we will address current challenges of privacy in machine learning, including memorization issues, inference privacy violations and methodologies for text anonymization. Following, we will discuss the current scientific and regulatory approach to privacy in Large Language Models; emerging Inferential privacy threats; and try to define the balance between data utility and privacy protection. Additionally, the course day examines content ownership challenges, such as fake news and the use of watermarks to verify AI-generated content alongside their ethical considerations. Participants will gain a deeper understanding of contemporary privacy challenges in LLMs, ensuring they are well-informed and knowledgeable.

Course Format

lecture, hands-on workshops, group discussions & demonstrations

Day Schedule

Time Slot	Lecture/Workshop Title	Key Topics Covered	Responsible
09:00 – 10:00	**Presentation of Student Case Studies on Research Integrity**	Compromising confidentiality in the peer review process	MP, BG
		The notions of scientific authorship in the era of generative AI	
		R.I.P. scientific publishing and peer-review system	
		Automation in research, parroting, epistemic risks, generative AI] Automation of the research cycle: epidemic opportunity or risk?	
10:00-11:00	**Input**: Secure Handling of Personal and Confidential Research Data	ETH Regulations & examples	FO, FS
11:00-11:10	Break		
11:10-11:40	**Lecture 1:** Introduction to Privacy	Historical handling of privacy and personal data; Privacy Regulations in ML, EU AI Act	RS
11:40-13:00	**Workshop 1**: Memorization	Scientific Notions of Privacy for LLMs, Memorization, Inferential Privacy	RS
	Discussion: Privacy risks associated with memorization	Practical solutions and criticisms; What is personal data in these	RS

		cases? Are current definitions compatible with the field?	
13:00-14:00	Lunch Break		
14:00-15:00	**Lecture & Workshop 2**: Anonymization	Applying anonymization tools to text; Attempting re-identification of anonymized data	RS
	Discussion: Effectiveness of current methods for Anonymization	Effectiveness of current methods; Utility-privacy trade-offs	RS
15:00-15:30	**Lecture 3**: Watermarking	Human vs AI generated texts; Detection techniques for AI-generated content; Watermarking	RS
	Discussion: Effectiveness and uses of watermarking	Future implications in content verification	RS
15:30-16:30	**Workshop 3**: Watermarking	Identifying AI generated text. Trying to come up with text that is detected as AI generated or as not AI generated; Identifying and manipulating watermarked text; Games and exercises on watermark application	RS
16:30-17:00	Discussion + Buffer + Closing	Summary of the Day	RS
17:00-17:30	**Presentation of Student Case Studies on Research Integrity**	Fair Use, Copyright, generative AI] Data Feasts and Fair Use: AI's Legal Appetite Under Scrutiny	MP, BG
		[Human in the loop, generative AI] The Human Paradox: Objective Instruments, Subjective Minds - Revealing the Illusion of Precision: Between Machine Accuracy and Human Judgment	
17:30-17:45	Course Closing Statements		BG

Learning Objectives
The goal for all participants is to understand the current privacy implications of LLMs. In particular, they will learn about the main state of privacy research for LLMs having reproduced/observed (on a small scale) some of the most important findings of the last years. Further they will know about language model watermarking and how to apply it to LLM generated texts in order to detect it later.
Lectures and Workshop Details

Input – Secure Handling of Personal and Confidential Research Data - ETH Regulations and Guidelines. This session will provide basic information on ETH regulations that apply for handling of research data and code along the research data life cycle. We will raise awareness for data privacy, data protection and cybersecurity. The ETH's classification system defining data confidentiality will be introduced, which allows data owners to assess the level of protection required for a specific type of data. We will also present examples from various disciplines where research data requires special protection and present do's and don'ts in dealing with sensitive research data. Finally, we will highlight specific tools for data sharing and collaborative work that are available at ETH and will mention the confidentiality class for which these tools have been approved.

Lecture 1 - Privacy Regulations in ML and the EU AI Act. This lecture provides an introduction to key privacy regulations affecting machine learning, with a focus on the GDPR, CCPA, and the EU AI Act. Participants will gain an understanding of the historical context of data privacy, the current regulatory landscape, and upcoming changes in legislation. We will also discuss the challenges and ambiguities present in the new AI Act and its impact on AI practices.

Workshop 1 - Scientific Notions of Privacy for LLMs: Memorization & Inference. This workshop explores the concept of memorization in large language models (LLMs) and its implications for privacy. Participants will engage in hands-on exercises to extract memorized data from models and discuss the privacy risks associated with memorization. The session will cover scaling laws, privacy aspects covered by memorization, and practical methods to address these challenges. Further we will investigate inference-based privacy violations not covered by memorization, highlighting potential future work in privacy research. Requirements: this workshop requires access to a browser and an active Google account for Google Colab.

Lecture & Workshop 2 - Anonymization of Text. This session focuses on the techniques and challenges of text anonymization. Participants will experiment with current anonymization tools to assess their effectiveness and explore the trade-offs between data utility and privacy. The workshop includes interactive exercises on anonymizing text and attempts to re-identify anonymized data, followed by discussions on improving anonymization methods. Requirements: Requires access to a browser and a Google account for Google Colab.

Lecture 3 - LLM Watermarking and Detection of AI-Generated Text. This lecture addresses the issues of content ownership and authenticity in the age of AI-generated text. We will explore the concept of watermarking in language models, discussing its potential to combat fake news and enhance content credibility. Similarly we will deal with on-going issues trying to verify whether a text is actually human-generated. The session will cover the pros and cons of watermarking, robustness, and the challenges of detecting AI-generated text.

Workshop 3 – Watermarking. In this session, participants will engage in activities designed to explore the practical application of watermarking in AI-generated content. Through several guided exercises, attendees will learn to identify and manipulate watermarked text, gaining insights into the effectiveness, risks, and potential uses of LLM watermarking. The session concludes with a discussion on the role of watermarks in content verification. Requirements: Requires access to a browser and a Google account for Google Colab.

Materials and Resources
The workshop will require access to a browser and a Google account for Google Colab.
Bring your own laptops